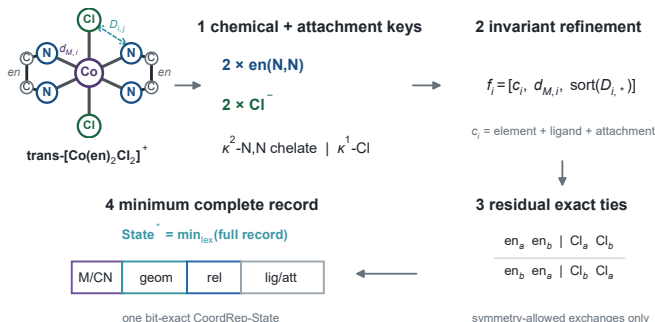


(A) Attachment-aware canonicalization



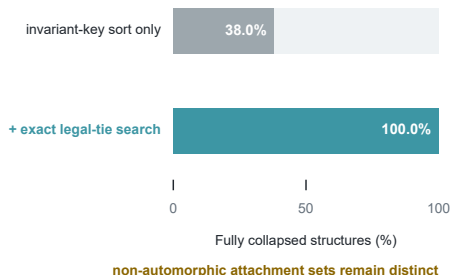
(B) End-to-end invariance challenge

$N = 1,000; K = 100$ per arm

	Rigid	Reindexed	Both
ordered Cartesian	0.0%	0.0%	0.0%
ordered distance matrix	100.0%	0.0%	0.0%
relation-stripped key	100.0%	100.0%	100.0%
same fields, source order	99.7%	5.0%	5.0%
CoordRep-State	100.0%	100.0%	100.0%

(C) Exact ties require whole-record minimization

real orbit cohort: 1,000/5,173; $K = 100$ reindexings



(D) Donor relations preserve stereochemical specificity

shared control: $\text{M/CN} = \text{Co(III)/6} \mid \text{geom} = \text{O}_h \mid \text{lig/att} = 3 \times \kappa^1\text{-NH}_3 + 3 \times \kappa^1\text{-Cl}$

